Week 5 — Random Numbers & Rock-Paper-Scissors

Name: ____ Date: ____

> `import random` *borrows a* ***toolbox****. Its tools give every outcome an* ***equal chance****.*

Today's words

Word	What it means
`import`	borrows a toolbox (module) of ready-made tools
`random.randint(a, b)`	a whole number from a to b — **both ends included**
`random.choice([...])`	picks ONE item from a collection in square brackets
equally likely	every outcome has the same chance
fair	no player (and no outcome) has an edge

1 · Prove the game is fair 🪨📄✂️

Fill every box with **W** (class wins), **L** (class loses), or **T** (tie). Remember: rock smashes scissors · paper wraps rock · scissors cut paper.

CLASS throws ↓ · COMPUTER throws →	rock	paper	scissors
rock			
paper			
scissors			

Count your letters: W: _ ***L:*** _ T: ____ → Is the game fair? **YES / NO**

2 · Which spinner is fair? 🎡

Three computer "spinners." **Circle the fair one** (all outcomes equally likely):

```
A)  random.choice(["red", "red", "blue"])
B)  random.choice(["red", "blue", "green"])
C)  random.choice(["win", "win", "win", "lose"])
```

In spinner A, the chance of red is _ ***out of*** *. In spinner C, the chance of lose is _* ***out of*** .

3 · What's possible? 🎲

Circle **every** number that `random.randint(3, 7)` could give:

> *0 1 2 3 4 5 6 7 8 9*

🧠 Brain teaser (optional — take it home)

A sneaky computer plays rock-paper-scissors, but it picks its throw with:

```
random.choice(["rock", "rock", "paper"])
```

You get to pick ONE throw and stick with it. **Which throw gives you the best chance — and what is your chance of winning? Can you ever lose?** (Hint: list the 3 things the computer might do and what happens to your throw in each.) Bring your answer next week for a shout-out.